

A-4. Introduction of Climate-resilient Varieties

 Agriculture Land Conservation	 High	 Medium		
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions									
Measures	The introduction of climate-resilient crop varieties is a crucial strategy for adapting to the impacts of climate change on agriculture. Climate-resilient varieties are specifically developed to withstand the global temperatures rise, unpredictable rainfall patterns, and the spread of pests and diseases. They are bred to thrive under stress conditions such as drought, heat, salinity, or flooding. These varieties can be genetically modified or developed through conventional breeding methods, such as hybridization, cross-breeding, or selection of native traits that exhibit greater resilience.								
	Introducing climate-resilient crop varieties is an effective adaptation measure that help the agriculture dependent community in CHTs cope with the adverse effects of climate change. By improving the ability of crops to withstand environmental stresses, it can be crucial in securing food production, enhancing food security, and supporting the resilience of farming communities.								
Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
	■	■	■	■	■	■	■	■	■
Applicable Sites	Suitable for degraded lands, barren area, low forest								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
Design options and performance	• Drought-Resistant Crop Varieties – Includes drought-tolerant rice, maize, sorghum, and millets for water-scarce conditions. • Flood-Resilient & Waterlogging-Tolerant Crops – Includes submergence-resistant rice varieties like BRRI dhan51 and BRRI dhan52. • High-Temperature Tolerant Crops – Includes heat-resistant wheat, maize, and climate-adapted vegetable hybrids. • Erosion-Preventing Leguminous Crops – Deep-rooted legumes (cowpeas, pigeon peas) for slope stabilization and nitrogen enrichment. • Agroforestry-Based Resilient Farming – Integrates multi-strata agroforestry with stress-resistant fruit trees and crops.								
Feasibility criteria	• Choosing crop varieties suited to the specific climatic stresses of the region, such as drought, heat, flooding, or salinity, ensuring they thrive under local environmental conditions. • Considering the cost of seeds, production, and maintenance for farmers, particularly smallholders, and the associated return on investment despite potential climate-related challenges.								

	Choosing crop species that are resistant to emerging pests and diseases to reduce the need for chemical interventions.			
Operation and maintenance	The operation and maintenance challenges of climate-resilient crop varieties include high initial costs, limited seed availability, and the need for specialized knowledge and management practices, such as pest control and soil fertility.			
Cost	High			
Benefits				
	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			

	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	• Climate-resilient varieties, especially drought-tolerant crops, require less water to grow hence help conserve freshwater resources, particularly in water-scarce regions. • Some climate-resilient crops with deeper root systems help anchor soil, reduce erosion, and improve soil structure thus maintain soil fertility and protect agricultural land from degradation. • Climate-resilient crop varieties form diversified farming systems, support greater biodiversity, encourage the presence of beneficial insects, pollinators, and soil organisms, which contribute to overall ecosystem health. • Climate-resilient crop varieties with improved root systems, organic matter from plants, and healthy soil management practices help absorb and store carbon in the soil, acting as a natural carbon sink. • Climate-resilient crop varieties usually have lower pesticide and fertilizer requirement than other hence help conserve the natural ecosystem.			
Remarks	This measure will aid landscape restoration and increase agricultural production in CHT of Bangladesh.			